

Midterm 2 Part b

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OlcaY Taner YILDIZ

I. QUESTION (12 POINTS)

Given the following error rates for a four feature problem. Error Rates: (F_1) , 0.23; (F_2) , 0.15; (F_3) , 0.16; (F_4) , 0.24; (F_1, F_2) , 0.18; (F_1, F_3) , 0.14; (F_1, F_4) , 0.20; (F_2, F_3) , 0.16; (F_2, F_4) , 0.29; (F_3, F_4) , 0.22; (F_1, F_2, F_3) , 0.19; (F_1, F_2, F_4) , 0.25; (F_1, F_3, F_4) , 0.24; (F_2, F_3, F_4) , 0.26; (F_1, F_2, F_3, F_4) , 0.29.

- What will be the result of the stepwise forward selection?
- What will be the result of the stepwise backward selection?
- What will be the result of the floating search?
- What is the optimal subset?

II. QUESTION (10 POINTS)

Show the steps taken by the decision tree algorithm according to the following training examples.

Class	a_1	a_2
+	T	T
+	T	T
-	T	F
+	F	F
-	F	T
-	F	T

$$\log_2 \frac{2}{3} = -0.585, \log_2 \frac{1}{3} = -1.585, \log_2 \frac{1}{2} = -1.$$

Draw the decision tree learned for this dataset.

III. QUESTION (8 POINTS)

Draw the linear perceptron for regression problem. Derive the update equations. Remember the error function is

$$E = \sum_t (r^{(t)} - y^{(t)})^2 \quad (1)$$

and $y^{(t)}$ is defined as

$$y^{(t)} = \sum_i w_i x_i^{(t)} + w_0 \quad (2)$$

where w_i 's and w_0 are the weights .

IV. QUESTION (9 POINTS)

Suppose that the task is to cluster the following eight points (with (x, y) representing location) into three clusters. $A_1(2, 10)$, $A_2(2, 5)$, $A_3(8, 4)$, $A_4(5, 8)$, $A_5(7, 5)$, $A_6(6, 4)$, $A_7(1, 2)$, $A_8(4, 9)$. The distance function is Euclidian distance. Suppose initially we assign A_2 , A_5 , and A_8 as the center of each cluster, respectively. Run the k-means algorithm one epoch on this data.